Algorithmic Geometry Fall, 2026

TD 3

1. Give a complete proof of $O(\sqrt{n \ln m})$ bound for discrepancy on a set system with n elements and m sets.

   Hint: You may find the following version of Chernoff's bound useful:

   Lemma 0.1. *[Chernoff's bound] Let $X_1, \ldots, X_s$ be independent random variables with $X_i \in \{-1, 1\}$ and $\Pr[X_i = -1] = \Pr[X_i = 1] = 1/2$, for $i = 1, \ldots s$. Set $X = \sum_{i=1}^{s} X_i$. Then, for any $\Delta > 0$, we have*

   $$\Pr[|X| \geq \Delta] \leq 2e^{-\Delta^2/2s}.$$

   Solution. First, we consider the situation for a single set $S \in \mathcal{R}$. A simple calculation shows that *in expectation*, the random coloring χ of S is perfectly balanced:

   $$\mathrm{E}_\chi\left[\sum_{v \in S} \chi(v)\right] = \sum_{v \in S} \mathrm{E}_\chi\left[\chi(v)\right] = \sum_{v \in S}\left(\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot (-1)\right) = 0.$$

   However, we need the fact that χ is not only balanced in expectation, but *with high probability*, simultaneously for all sets. For this, we need a *concentration bound* which shows that the value of a random variable is close to its expected value, with high probability. In our setting, the most applicable concentration bound is *Chernoff's bound*. Using the version of Chernoff's bound that is given in the hint, we determine a $\Delta > 0$ such that the probability

   $$\Pr_\chi\left[\mathrm{disc}_\chi(S) \geq \Delta\right] = \Pr_\chi\left[\left|\sum_{v \in S} \chi(v)\right| \geq \Delta\right]$$

   is small. For this, let $S = \{v_1, \ldots, v_s\}$, $s = |S|$, and set $X_i = \chi(v_i)$, for $i = 1, \ldots, s$. Then, we have $X_i \in \{-1, 1\}$, and $\Pr[X_i = -1] = \Pr[X_i = 1] = 1/2$, independently of the others. We set $\Delta = \sqrt{2s \ln(2m^2)}$. Then, using Lemma 0.1 (page 1), we have

   $$\begin{aligned}\Pr_\chi\left[\left|\sum_{v \in S} \chi(v)\right| \geq \sqrt{2s \ln(2m^2)}\right] &= \Pr_\chi\left[\left|\sum_{i=1}^{s} X_i\right| \geq \sqrt{2s \ln(2m^2)}\right] \\ &\leq 2e^{-(2s \ln(2m^2))/2s} = 2e^{-\ln(2m^2)} = \frac{1}{m^2}.\end{aligned}$$

   Now, having analyzed the situation for a single set, let us bound the probability that there *exists* a set with discrepancy at least $\sqrt{2n \ln(2m^2)}$:

   $$\Pr_\chi\left[\text{there exists an } S \in \mathcal{R} \text{ with } \mathrm{disc}_\chi(S) \geq \sqrt{2n \ln(2m^2)}\right] \leq \sum_{S \in \mathcal{R}} \Pr_\chi\left[\mathrm{disc}_\chi(S) \geq \sqrt{2n \ln(2m^2)}\right]$$

   $$\leq \sum_{S \in \mathcal{R}} \Pr_\chi\left[\mathrm{disc}_\chi(S) \geq \sqrt{2|S| \ln(2m^2)}\right] \leq \sum_{S \in \mathcal{R}} \frac{1}{m^2} = \frac{1}{m}.$$

   Thus, we have

   $$\Pr_\chi\left[\text{for all } S \in \mathcal{R}, \text{ we have } \mathrm{disc}_\chi(S) < \sqrt{2n \ln(2m^2)}\right] \geq 1 - \frac{1}{m}.$$

   The claim follows, noting that $\sqrt{2n \ln(2m^2)} = O\left(\sqrt{n \ln m}\right)$.

2. Given a finite set system $(X, \mathcal{R})$ and a $\epsilon > 0$, show that a uniform random sample of size $\Omega\left(\frac{\ln m}{\epsilon^2}\right)$ is an ϵ-approximation of $\mathcal{R}$ with high probability.

Solution. Construct a random multiset A of size a by picking independently an element of X uniformly at random, a times. Fix a $S \in \mathcal{R}$, and set the indicator random variable X_i to 1 iff the i-th element of A lies in S. Then

$$\mathrm{E}\left[\sum_{i\in[a]} X_i\right] = a \cdot \frac{|S|}{|X|}.$$

Applying Chernoff bound from CM3,

$$\Pr\left[|S \cap A| > a \cdot \frac{|S|}{|X|} + \epsilon a\right] + \Pr\left[|S \cap A| < a \cdot \frac{|S|}{|X|} - \epsilon a\right] \leq 2\exp\left(-\frac{2\epsilon^2 a^2}{a}\right) = 2\exp\left(-2\epsilon^2 a\right).$$

Setting $a = \Theta\left(\frac{\ln m}{\epsilon^2}\right)$ makes this probability $O(1/m^2)$; now by the union bound, with probability at least $1 - 1/m$, the random sample A is an ϵ-approximation for $\mathcal{R}$.